HEALTHCARE AI PROJECT | IBM Z DATATHON 2026

AI-Powered Real-Time Patient Deterioration & Emergency Decision Support System

TEAM DETAILS

TEAM NAME: **ZYNC**

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Problem & Solution

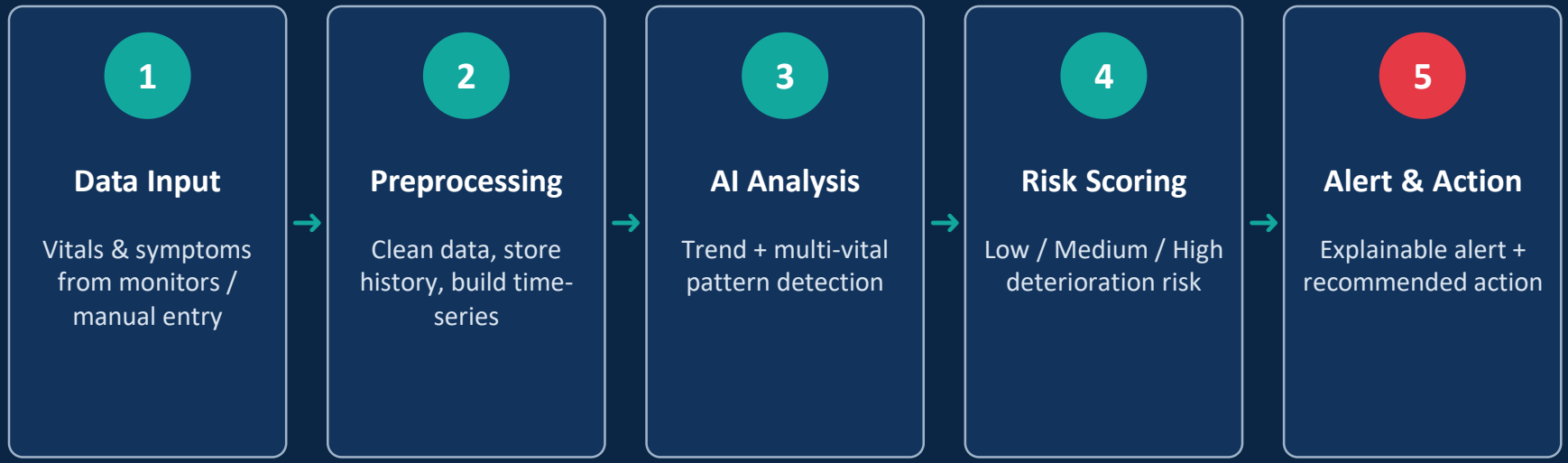
THE PROBLEM

- Patient deterioration is often noticed too late
- Alerts trigger only on a single threshold (e.g. SpO_2 = 88%)
- Trends and combined vital changes are missed
- Doctors & nurses face heavy workload and alert fatigue

OUR SOLUTION

- AI monitors HR, SpO_2, BP, Temp, Resp. Rate & symptoms in real time
- Analyzes current values + history + trends together
- Gives an early warning with Risk Level (Low / Medium / High)
- Explains WHY the alert was raised and recommends action

Working Flow



Doctor / nurse reviews the alert and takes the final clinical decision.

Applications

ICU & Critical Care

Continuous early warning for high-risk patients

Hospital Wards

Helps nurses prioritize which patient needs attention first

Emergency Department

Faster triage and urgent-case identification

Remote / Home Monitoring

Track post-discharge and chronic patients with wearables

Rural & Small Clinics

Decision support where specialists are limited

Ambulance / Pre-hospital

Early alert sent to hospital before arrival

Example: Trend-Based Early Detection



HIGH RISK ALERT

Risk Level: HIGH

Reason: SpO_2 continuously decreasing while heart rate increasing – deterioration trend in respiratory & cardiovascular parameters.

Action: Immediate clinical assessment required.

Key Features & Benefits

Trend-Based Detection

Looks at how vitals change over time, not just one reading

Multi-Parameter Analysis

Combines HR, SpO_2, BP, Temp, RR and symptoms

Explainable Alerts

Clear reason shown with every alert – builds doctor trust

Risk Prioritization

Ranks patients so urgent cases are seen first

Reduced Alert Fatigue

Fewer false alarms than single-threshold systems

Faster Decisions

Early warning gives clinicians more time to act

Technology Stack: Platform & ML Engine

IBM Z Platform

- IBM z16 / LinuxONE (s390x)
- Linux on IBM Z virtual server
- IBM Integrated Accelerator for AI: on-chip, low-latency scoring
- Pervasive hardware encryption for PHI / HIPAA
- 99.99999% mainframe availability

Runtime Environment

- Python 3.12
- Jupyter Notebook as self-contained backend and live bedside demo
- In-memory deterministic processing
- Single-threaded (n_jobs=1): no disk I/O or subprocess overhead

ML Engine

- LightGBM (LGBMClassifier): 180 trees, lr 0.03, depth 5, 24 leaves, subsample 0.85 / 0.80
- GroupShuffleSplit grouped by parent_id: no clone data leakage
- Tested on 5,505 real ICU admissions
- AUROC, AUPRC, Balanced Accuracy

Technology Stack: Features, XAI & Visuals

Data & Feature Engineering

- Pandas & NumPy; hourly EHR telemetry
- ffill().bfill() per patient + median imputation
- Shock Index (HR / SBP)
- Modified Shock Index (HR / MAP)
- Hypoxia-Tachycardia Divergence (HR $\Delta > 4$ bpm & SpO_2 $\Delta < -2\%$)
- Missingness Acuity (total_labs_drawn)

Explainable AI

- TreeSHAP (shap.TreeExplainer)
- Game-theoretic feature contributions
- Bedside view: WBC surge, Creatinine rise, Shock Index spike
- Global feature importance

Visualization

- Matplotlib & Seaborn
- ROC & PR curves (dual panel)
- Global SHAP bar chart
- 6-hour trajectory: HR, MAP, SpO_2, Resp. Rate

Dataset: PhysioNet / Computing in Cardiology Challenge 2019 (Early Detection of Sepsis), augmented for class balance

Future Scope & Conclusion

FUTURE SCOPE

- Integration with hospital EHR / wearable devices
- Train on larger real clinical datasets
- Mobile alerts for doctors and nurses
- Multilingual dashboard for rural healthcare

CONCLUSION

Our system acts as an early-warning and decision-support layer that helps detect critical patients before their condition becomes severe.

⚠ Supports doctors & nurses – does not replace clinical judgment or give autonomous diagnosis.

Thank You

Questions & Feedback are welcome

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